CAMPORA - A Unified Smart Campus Ordering and Service Platform

1. Project Overview

Campora is a unified smart campus ordering and service management platform developed to digitize multiple campus services through a single mobile application and web-based management system. The platform enables students and faculty to order food, purchase stationery and books, upload documents for printing, make digital payments through a simulated payment system, and collect orders using a token-based pickup process. Each department has a dedicated web portal to manage orders efficiently, while administrators can monitor overall operations through a centralized dashboard with analytics. Campora also includes an Agentic AI Voice Assistant that allows users to place orders and navigate the application using voice commands.

2. Problem Statement

Most college campuses still rely on manual ordering and queue-based service management. Students must visit multiple counters to purchase food, stationery, books, or submit printing requests, leading to long waiting times, inefficient service management, and lack of centralized monitoring. Existing systems generally focus on only one service and do not provide an integrated solution for campus operations.

3. Proposed Solution

Campora provides a single digital platform where users can access multiple campus services through one application. The system simplifies ordering, payment, token generation, and service management while improving efficiency for both students and campus departments. It eliminates unnecessary queues and provides a centralized administrative platform for monitoring all campus services.

4. Project Objectives

- Develop a unified mobile application for campus services.
- Reduce waiting time through online ordering and token-based collection.
- Simplify food, stationery, book, and printing services.
- Provide separate management portals for each department.
- Develop an administrator dashboard for analytics and reporting.

- Integrate an Agentic AI Voice Assistant for voice-based interaction.
- Create a scalable and user-friendly campus management system.

5. Project Modules

Student Mobile Application

- User Registration
- User Login
- Home Dashboard
- Food Ordering
- Stationery Purchase
- Book Purchase
- Print Service
- Shopping Cart
- Simulated Payment
- Token Generation
- Notifications
- Profile Management
- AI Voice Assistant

Food Court Portal

- Login
- View Orders
- Accept Orders
- Update Order Status
- Mark Ready
- Complete Delivery

Stationery Portal

- Product Management
- Receive Orders
- Inventory Management
- Order Completion

Book Store Portal

- Manage Books
- Receive Orders
- Update Availability
- Complete Orders

Print Centre Portal

- Receive Print Requests
- Download Uploaded Documents
- Print Management
- Update Status

Admin Dashboard

- Dashboard
- User Management
- Department Management
- Revenue Analytics
- Order Reports
- Inventory Monitoring

6. Technologies Used

Technology	Purpose
React.js	Frontend Development
Bootstrap	Responsive User Interface
Node.js	Backend Runtime Environment
Express.js	REST API Development
MongoDB	Database
MongoDB Compass	Database Management
Postman	API Testing
Git	Version Control
GitHub	Code Repository
VS Code	Development Environment

7. Database Collections

- Users
- FoodItems
- StationeryItems
- Books
- PrintOrders
- Orders
- Payments
- Tokens
- Notifications
- Departments

8. User Roles

Student

- Register
- Login
- Order Food
- Purchase Stationery
- Purchase Books
- Upload Print Documents
- Make Payment
- Receive Token
- View Notifications

Department Staff

- Login
- View Orders
- Accept Orders
- Update Status
- Complete Orders

Administrator

- Monitor All Departments
- Manage Users
- Generate Reports
- View Analytics
- Manage Inventory

9. Key Features

- Unified Campus Platform
- Food Ordering
- Stationery Purchase
- Book Purchase
- Print Service
- Simulated Payment
- Token-Based Order Collection
- Department Portals
- Admin Dashboard
- Analytics
- Agentic AI Voice Assistant
- Secure Authentication
- Responsive User Interface

10. Expected Outcome

The proposed system will provide a smart, centralized platform for managing essential campus services efficiently. Campora will reduce waiting time, simplify service access, improve departmental coordination, and enhance the overall campus experience through digital transformation.

11. Future Scope

- Real Payment Gateway Integration
- QR Code-Based Order Verification
- Push Notifications
- College ERP Integration
- Attendance Integration
- AI-Based Recommendations
- Android and iOS Deployment

12. Project Timeline

Phase	Duration
Requirement Analysis	July
UI/UX Design	July – August
Frontend Development	August – October
Backend Development	October – November
Database Integration	November
AI Integration	December
Testing	January
Documentation	February
Final Deployment & Review	March